Zilionis 2019 NSCLC — analysis summary

RUN 2 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6/7 samples (≥131 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 2.85 of 6); proliferation globally low (6.4% cycling, <12%/lobe) — no cycling compartment to scope out.

Gate 2 — module geography → GO (non-branching)

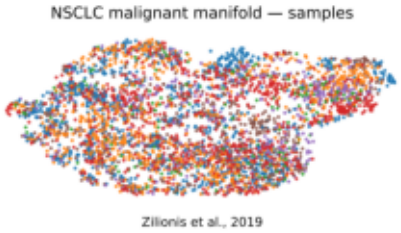
Single Gast(early)→pSom(late) cross-repression axis: Gast falls along pseudotime (ρ=−0.58), pSom rises (ρ=+0.74); smoothed Gast↔pSom r=−0.53. Crest a mid gradient (leans pSom side, r=−0.36 vs −0.51). No Gast/pSom co-expression hub — collapsed bifurcation.

Gate 3 — hub mutual exclusivity → NA

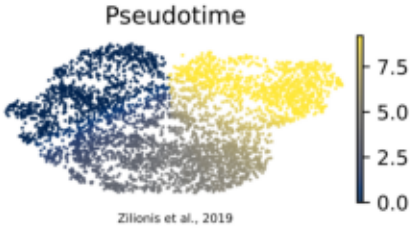
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

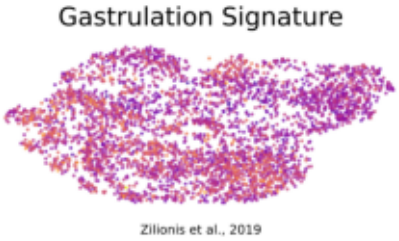
Manifold — samples



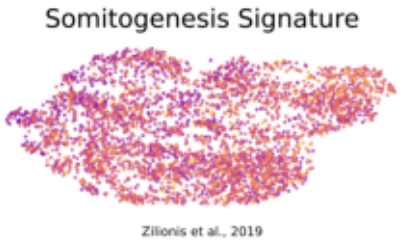
Pseudotime



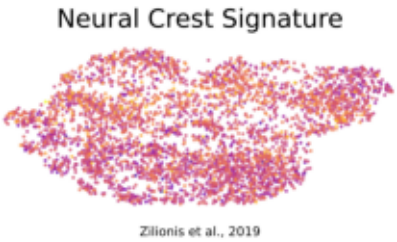
Gastrulation



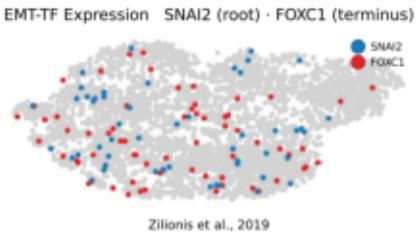
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Non-Small Cell Lung Cancer (Zilionis et al., 2019), inDrop — 7 patients / 31,179 cells (master scope); malignant only: 3,995 load, 3,920 across 6 samples (≥131) enter preprocessing; whole malignant manifold (no lobe scoping).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Zilionis NSCLC malignant manifold is a single monotone gastrulation → posterior-somitogenesis axis (Gast-high origin → pSom-high terminus), matching the developmental order; it lacks Puram's hub-and-spoke bifurcation (PPT tree hairballs, no Gast/pSom co-expression hub), so Gate 3 has no hub to test. The developmental axis spans low-cycling lobes, so the whole manifold is kept (Bischoff/Sun no-scoping precedent). EMT-TFs SNAI2 (early/root) → FOXC1 (late/terminus); canonical ZEB1/TWIST1 near the detection floor (1.5%/0.2%); EMT-TF signal weak, end-separation negligible (Δpt 0.03).

Parameter choices

Min cells/sample	131 (keeps 6/7; gap 232→75)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = Gast-high, Crest-low	**Geometry**	Linear-pSom/Crest (non-branching)